

EMILIO FONSECA

Full-Stack Developer

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SUMMARY

Full-Stack Developer who builds production software end to end — web apps (ERP, e-commerce, management tools), data & BI, and applied AI. I integrate REST APIs with Python (FastAPI) and Node.js and turn manual operations into software that runs itself. Currently shipping to production; portfolio with live, explorable demos available.

EXPERIENCE

Full-Stack Developer — MN del Golfo

2025 – Present

Materials retail chain · Tamaulipas, MX

- Ship internal production software — web apps, dashboards and automations — used daily across the company's branches, replacing manual workflows with self-running tools.
- Built AI assistants on the official WhatsApp Business API (text, voice and image) that resolve routine queries 24/7 and cut hours of repetitive manual work each week.
- Integrated systems end to end with Node.js, React, SQL Server, Firebase and Docker — including the ERP (Dynamics 365 Business Central) — so data flows automatically instead of by hand.

PROJECTS

Dexter — Open-source AI agent platform

Apache 2.0

Python · Go (TUI) · SQLite (FTS5) · MCP · multi-LLM

- Built a configurator that extends AI agents with 46+ lazy-loaded skills and persistent memory (Engram, SQLite FTS5) across 6 agent platforms, backed by 99 automated tests.
- Implemented a multi-LLM router with fallback across 4 providers (Anthropic, OpenAI, Google, Ollama), a WhatsApp auto-responder, computer-use GUI control and 112+ importable community flows.

Vértice — Full-stack work-management app

Live demo

React · TypeScript · FastAPI · PostgreSQL · Docker · offline-first PWA

- Developed an offline-first PWA to manage projects, quotes and payments end to end, with secure JWT auth and a Dockerized FastAPI + PostgreSQL backend.

Vitrina — E-commerce storefront

Live demo

React · Vite · responsive UI

- Built a responsive online store with category catalog, cart with promotions, stock alerts and instant product search.

Dino AI — Reinforcement-learning game agent

GitHub

Python · Stable-Baselines3 (PPO) · OpenCV · PyTorch

- Trained a reinforcement-learning agent (PPO) that learns to play Chrome's dino game on its own from on-screen pixels via computer vision (OpenCV).

EDUCATION

B.Sc. in Computer Systems Engineering

2020 – 2025

Universidad Autónoma de Tamaulipas

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C# · **Frontend:** React, TypeScript, Vite, Tailwind, PWA / offline-first · **Backend:** Node.js, Express, FastAPI, REST, JWT, OAuth · **Data & DevOps:** PostgreSQL, SQL Server, Firebase, Docker, Git, Linux, CI/CD · **AI & LLMs:** GPT-4o, Claude, Whisper, RAG, prompt engineering, tool calling, MCP

CERTIFICATIONS

AI from Zero to Production — BIG School (2026); Data Fundamentals — Google / Coursera (2026); additional Coursera credentials, and Mastermind certificates in development, cybersecurity and game development.

HONORS & LANGUAGES

ICPC Mexico Grand Prix 2022 — competitive algorithmic problem solving. · **Languages:** Spanish (Native), English (B2, Intermediate-Advanced).